

to address affordability of the devices. An online free resource called DAKSH that caters to the computer skills training and special skills related to ASD has been hosted at <http://www.learn4autism.com/>. This also allows people to collaborate and share content, specific to the field of autism, with each other. We aim to make this the largest free content resource for the field in the coming years. There are a few other initiatives that are at a very early stage focusing on other special needs."

Automated prescription reader for visually-challenged patients

En-Vision's ScripTalk Station is an automated prescription reader for visually-impaired, dyslexic and illiterate users, who find it difficult to read the label information on prescription medication. It uses radio frequency identification (RFID), TTS technologies and Analog Devices' Blackfin processor to transform drug-label printing into audible, spoken words.

When a sight-impaired patient places a prescription bottle on the ScripTalk unit, a digitally-generated voice reads out the prescription label loud, clearly communicating both the drug name and the recommended dosage. As a result, people can manage their medical needs in a safe and private manner using a technology that is easy and affordable.

ScripTalk is a light-weight, portable, battery-powered device. Pharmacists connect the base station to a computer via a USB or serial port and upload prescription data from standard pharmacy-management software to the En-Vision device. With the press of a button, a special RFID-tagged label is encoded with all warnings, side effects and the patient's personal information. At home, the sight-impaired patient uses a similar device to hear the label information.

En-Vision partnered with Mistral Solutions, a concept-to-deployment design engineering firm based in India, to develop ScripTalk. They chose the Blackfin processor as it met all their needs in terms of price, power consumption, processing performance,



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peripheral interfacing and technology-ecosystem. Mistral was able to leverage its rich experience with Blackfin processors to enhance the ScripTalk Station's marketability by integrating multi-lingual TTS using Nuance RealSpeak for high-quality speech processing.

Narayanan Bhattathiripad, senior vice president-Product Engineering Services, Mistral Solutions, says, "ScripTalk was developed to cater to the needs of the visually-challenged. The solution, developed for the US-based En-Vision America, is an RFID tag based prescription reader. The reader eliminates dependency on human assistance for taking prescription medication. The prescription label information is programmed into RFID tags and pasted on drug containers used by patients. Patients use their own private ScripTalk Station Reader to read the label information, including medication name and dosage, instructions, warnings and other prescription details."

He also explains that containing the overall cost of the product was extremely critical as it dictated how well the product would be received in the market. Given its cost-effective core, Blackfin was found to be the best fit for this requirement.

Kantheti Srinivas, Processor and DSP Applications director, Analog Devices, explains, "One of the simplest ways to reduce the cost of a device is to combine

separate devices into one. With the programmable Blackfin processor, En-Vision found a single device that can simultaneously implement control functions and TTS processing that requires significant amount of DSP processing, reducing time-to-market, as well as bill-of-materials (BOM) and development costs. With Blackfin's on-chip power-management feature, designers were able to fine-tune the processor's power consumption profile without using an external power management IC."

As regards ScripTalk's USP, he says, "The product is lightweight and portable and works with any prescription in multiple languages. This works well with the visually-challenged, who have to refer to their prescriptions that are usually not embedded with the Braille script."

ScripTalk is used extensively in the US. Its developers have a tie-up with leading pharmacies like Walmart and CVS, providing ScripTalk an immense reach in the US market. The number of patient readers out in the US is now over 15,000. Today, patients obtain their reader for free, with a large pharmacy network partnering with En-Vision. Although the product is currently not available in India, En-Vision is fast expanding its markets globally.

Making music truly universal

Music is solace for the soul, for all. There have been brilliant examples in the past, of people with disabilities taking to music like fish to water. The famous solo percussionist, Dame Evelyn Glennie, for example, lost her hearing at the age of 12. Yet, she learnt to 'feel' the sound through sight, vibrations, etc, and went on to become one of the world's most renowned percussionists.

Thankfully, things are much easier today for people with disabilities who wish to learn and play music. Technology has helped prove that music is felt rather than simply heard—and if a special person has an innate talent, she can definitely bring it out for the world to see. There are, for example, apps that turn the iPad into touch-sensitive